Sectoral Segmentation using Clustering Techniques

Indian Financial Services Sector

1. Introduction

This study applies clustering techniques to segment firms in the Indian Financial Services sector using financial performance indicators. The objective is to identify groups of companies with similar profitability, liquidity, growth, and efficiency characteristics through unsupervised learning methods.

2. Dataset and Variable Selection

The dataset consists of 50 non-banking financial services companies selected after preprocessing, null-value elimination, and random filtering. Seven financial ratios were selected to capture different dimensions of company performance.

Financial Ratio	**Category**	**Purpose**
Return on Equity (ROE)	Profitability	Measures profitability generated from shareholder equity
Return on Assets (ROA)	Profitability	Measures efficiency of asset utilization
Current Ratio	Liquidity	Measures short-term solvency
Quick Ratio	Liquidity	Measures immediate liquidity
Revenue Growth (1 Year)	Growth	Measures yearly business expansion
Basic EPS	Profitability/Growth	Measures earnings generated per share
Total Asset Turnover	Efficiency	Measures operational efficiency

3. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed using histograms, boxplots, and correlation heatmaps to evaluate feature distributions, identify outliers, and understand relationships between variables.

The distribution plots indicate that liquidity variables such as Current Ratio and Quick Ratio are highly right-skewed due to a small number of firms with extremely high liquidity values. Basic EPS also shows strong skewness caused by a few firms with exceptionally large earnings per share.

Boxplot analysis confirmed the presence of multiple outliers across profitability, liquidity, and growth variables. Current Ratio and Quick Ratio displayed significant outliers, while Revenue Growth contained both highly positive and negative observations. These patterns indicate strong variability within the financial services sector.

Correlation analysis showed strong positive relationships between Return on Equity, Return on Assets, and Total Asset Turnover, suggesting that firms with higher profitability generally demonstrate better asset utilization efficiency. Current Ratio and Quick Ratio were almost perfectly correlated, which is expected because both measure liquidity performance.

4. Cluster Formation and Validation

K-Means clustering was applied to the scaled dataset after MinMax normalization. The Elbow Method was used to identify the optimal number of clusters by analyzing the reduction in within-cluster sum of squares across different K values.

The Elbow curve showed a noticeable bend at K = 3, indicating diminishing improvements beyond three clusters. Silhouette Score analysis further validated this result, with the highest silhouette score obtained at K = 3 (0.508). Therefore, three clusters were selected for final segmentation.

K Value	Silhouette Score	CH Score
2	0.457	23.44
3	0.508	29.62
4	0.453	24.22
5	0.453	26.52
6	0.369	27.74
7	0.367	29.25

5. Cluster Interpretation

Cluster-wise boxplot analysis revealed clear differences between the three groups.

- Cluster 0 represented diversified financial firms with moderate profitability, balanced liquidity ratios, and stable revenue growth. Most firms in this cluster displayed average operational efficiency and moderate EPS values.
- Cluster 1 consisted mainly of high-performing asset management and securities firms characterized by very high ROE, ROA, and Total Asset Turnover. These firms demonstrated superior operational efficiency and profitability compared to the other clusters.
- Cluster 2 represented investment-oriented firms with exceptionally high Current Ratio, Quick Ratio, and Basic EPS values. However, this cluster showed comparatively lower revenue growth and lower operational efficiency.

The cluster interpretation indicates that the Indian financial services sector contains structurally distinct business models with significant variation in profitability, liquidity, and growth characteristics.

6. Hierarchical Clustering

Hierarchical clustering using Ward linkage was also performed to validate the segmentation structure. The dendrogram and clustered heatmap showed meaningful grouping patterns among companies based on their financial characteristics.

Companies with strong profitability and operational efficiency were grouped together, while firms with extremely high liquidity and EPS values formed separate clusters. The hierarchical clustering results were broadly consistent with the K-Means clustering output.

7. Feature Importance Analysis

ANOVA F-score analysis was performed to identify the most influential variables contributing to cluster separation.

Feature	F-Score
Return on Assets	58.82
Return on Equity	37.56
Basic EPS	29.36
Total Asset Turnover	26.29
Current Ratio	25.45
Quick Ratio	24.97
Revenue Growth (1 Year)	14.34

Return on Assets emerged as the most influential variable, followed by Return on Equity and Basic EPS. This indicates that profitability and operational efficiency were the primary drivers of cluster formation.

8. Conclusion

The clustering analysis successfully segmented companies in the Indian Financial Services sector into three distinct groups based on financial performance indicators. The analysis demonstrated that firms differ substantially in profitability, liquidity, efficiency, and growth characteristics.

K-Means clustering with K = 3 provided interpretable and well-separated clusters, while hierarchical clustering further validated the segmentation structure. The study highlights the usefulness of unsupervised learning techniques for identifying meaningful patterns within financial datasets.